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| 5(a) | <p><i>both units must be used at least once</i></p> <p>any <b>three</b> from:</p> <ul style="list-style-type: none"> <li>1 no transpiration when LVPD is 0 kPa ;</li> <li>2 (both) increase in transpiration rate until 2.5 kPa ;</li> <li>3 <i>H. annuus</i> rate remains constant between 2.5 and 3.0 kPa, <i>N. oleander</i> rate decreases slightly ;</li> <li>4 <i>H. annuus</i> has higher rate of transpiration (at all LVPDs) ; <b>ora</b> for <i>N. oleander</i></li> <li>5 <i>H. annuus</i> has a higher rate of increase (between 0 and 2.5 kPa) ; <b>A</b> steeper gradient</li> <li>6 any comparative data quote from Fig. 5.1 ;</li> </ul> <p><i>must include figures and units from both axes <u>and</u> data must come from both species</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3 |
| 5(b) | <p><i>if a correct adaptation does not match the figure, allow ecf for the explanation Fig. 5.3</i></p> <p>any <b>four</b> from:</p> <ul style="list-style-type: none"> <li>1 three layers of thick-walled cells at, upper / lower, surface ; <ul style="list-style-type: none"> <li><b>A</b> two layers of cells under epidermis / epidermis and (two layers of) hypodermis</li> </ul> </li> <li>2 reduces, diffusion of water vapour to atmosphere / cuticular transpiration ; <ul style="list-style-type: none"> <li><b>A</b> increases the distance for water vapour to diffuse</li> </ul> </li> <li>3 no stomata on upper surface / stomata only on lower surface ;</li> <li>4 stomata not exposed to direct sunlight / AW ;</li> <li>5 (stomata in) pits / depressions / cavities / chambers / crypts / infoldings / grooves / AW (on lower side of leaf) ;</li> <li>6 creates humid atmosphere in, pit / AW <i>or</i> reduces the, diffusion / water potential, gradient (for water vapour) ; <ul style="list-style-type: none"> <li><b>A</b> vapour pressure gradient</li> <li><b>A</b> minimises effect of, external air currents / wind</li> </ul> </li> </ul> <p><i>Fig. 5.3 or Fig. 5.4</i></p> <ul style="list-style-type: none"> <li>7 thick walled epidermis ;</li> <li>8 reduces diffusion of water vapour through lower epidermis ;</li> <li>9 thick (waxy) cuticle (on epidermis / surface) ;</li> <li>10 reduces diffusion of water vapour through cuticle / reduces cuticular transpiration ; <b>A</b> (waxy) cuticle is waterproof</li> <li>11 (epidermal) hairs / trichomes (around stomata / fill pits) ;</li> <li>12 reduces air movement / traps still air / creates humid atmosphere / traps water vapour <i>or</i> reduces the, diffusion / water potential, gradient for water vapour ;</li> </ul> | 4 |